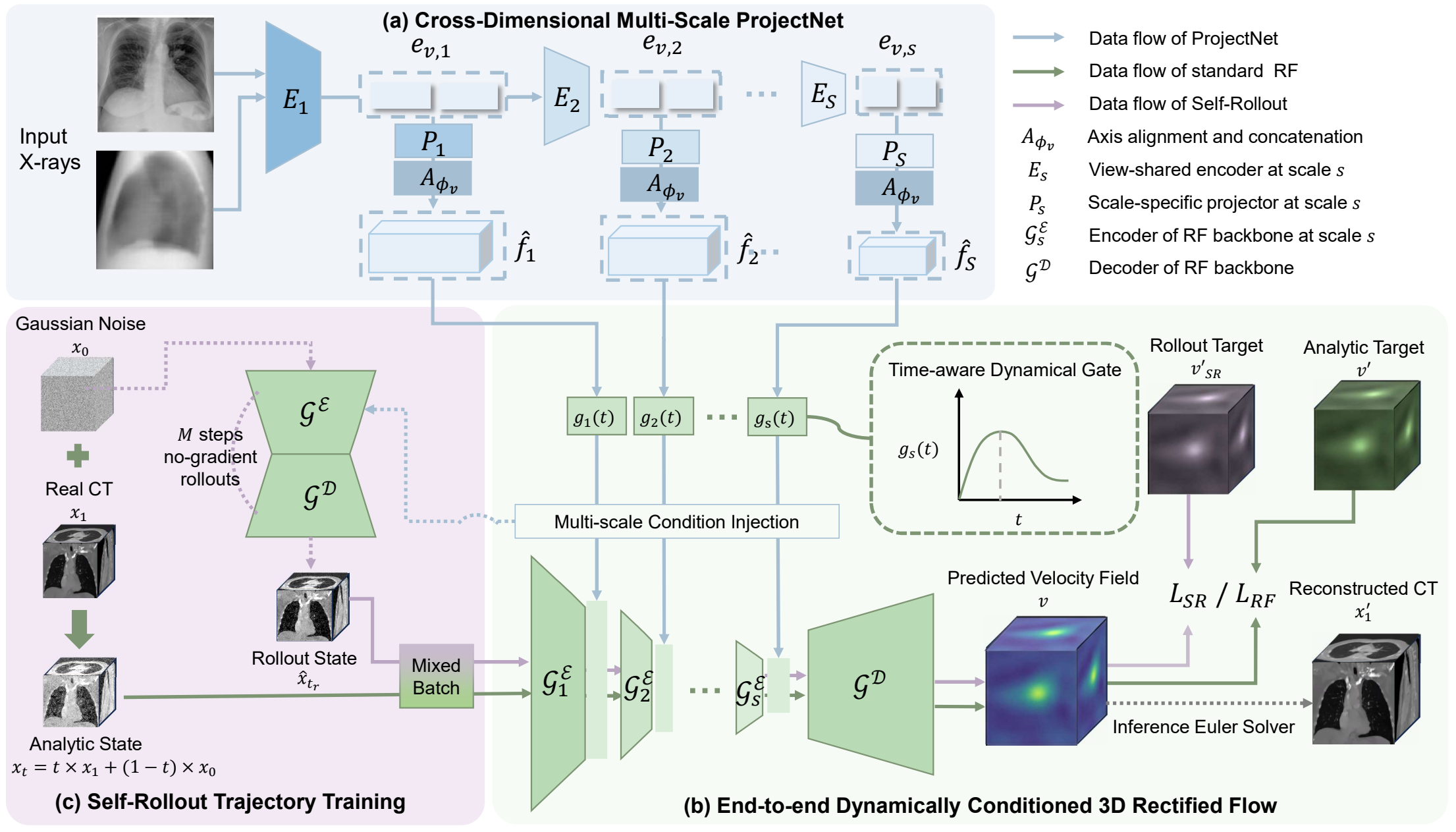

(a) Cross-Dimensional Multi-Scale ProjectNet
Input X-rays
E_1
E_2
E_S
$e_{v,1}$
$e_{v,2}$
$e_{v,s}$
P_1
P_2
P_S
A_{ϕ_v}
$\hat{f}_1$
$\hat{f}_2$
$\hat{f}_S$
Data flow of ProjectNet
Data flow of standard RF
Data flow of Self-Rollout
A_{ϕ_v} Axis alignment and concatenation
E_s View-shared encoder at scale s
P_s Scale-specific projector at scale s
$\mathcal{G}_s^{\mathcal{E}}$ Encoder of RF backbone at scale s
$\mathcal{G}^{\mathcal{D}}$ Decoder of RF backbone
Gaussian Noise
x_0
Real CT
x_1
M steps no-gradient rollouts
$\mathcal{G}^{\mathcal{E}}$
$\mathcal{G}^{\mathcal{D}}$
Rollout State
$\hat{x}_{t_r}$
Mixed Batch
Analytic State
$x_t = t \times x_1 + (1-t) \times x_0$
(c) Self-Rollout Trajectory Training
$g_1(t)$
$g_2(t)$
$g_s(t)$
Multi-scale Condition Injection
Time-aware Dynamical Gate
$g_s(t)$
t
Rollout Target
v'_{SR}
Analytic Target
v'
$\mathcal{G}_1^{\mathcal{E}}$
$\mathcal{G}_2^{\mathcal{E}}$
$\mathcal{G}_s^{\mathcal{E}}$
$\mathcal{G}^{\mathcal{D}}$
Predicted Velocity Field
v
L_{SR} / L_{RF}
Reconstructed CT
x'_1
Inference Euler Solver
(b) End-to-end Dynamically Conditioned 3D Rectified Flow